THE UK UNIVERSITY INTEGRATION BEE 2025/2026

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ONLINE ROUND

Sponsored by



Jane Street

Problems

1. $\displaystyle\int_0^{\frac{\pi}{4}} \frac{1}{x^2+1}\,\mathrm{d}x$
2. $\displaystyle\int_0^{\infty} \sin(\pi\,\lfloor x \rfloor)\,\mathrm{d}x$
3. $\displaystyle\int_0^{\infty} \operatorname{sech}(2\ln(x))\,\mathrm{d}x$
4. $\displaystyle\int_0^{a} e^{\frac{i\pi}{2}}\,\mathrm{d}z$
5. $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(x)}{1+\sqrt[x]{e}}\,\mathrm{d}x$
6. $\displaystyle\int_0^{1} \sqrt{-\ln(x)}\,\mathrm{d}x$
7. $\displaystyle\int_0^{1} \left(1+x+x^2+\cdots+x^{2025}\right) - \left(\{x\}+\{x\{x\}\}+\{x\{x\{x\}\}\}+\cdots+\underbrace{\{x\{\cdots\{x\}\cdots\}\}}_{\text{2025 times}}\right)\mathrm{d}x$
8. $\displaystyle\int_0^{\infty} \left(1-\frac{x}{1}\left(1-\frac{x}{2}\left(1-\frac{x}{3}\left(1-\frac{x}{4}\left(\cdots\right)\right)\right)\right)\right)\mathrm{d}x$
9. $\displaystyle\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{7x^7-5x^5+3x^3-x+1}{\cos^2(x)}\,\mathrm{d}x$
10. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\tan^{-1}(b\sin(x))}{\sin(x)}\,\mathrm{d}x$
11. $\displaystyle\int \frac{1}{x\sin^{-1}(\ln(x))\sqrt{1-(\ln(x))^2}}\,\mathrm{d}x$
12. $\displaystyle\lim_{n\to\infty}\int_0^{1} |x-|x^2-|x^3-|\cdots-|x^{n-1}-|x^n||\cdots||||\,\mathrm{d}x$
13. $\displaystyle\int_{|x|+|y|+|z|\le 1} \left\lceil x^2+y^2+z^2\right\rceil \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$
14. $\displaystyle\int_0^{2\pi} \sin(\sin(x)+2025x)\,\mathrm{d}x$
15. $\displaystyle\int_0^{\pi} \sqrt{1+\sin(2x)}\,\mathrm{d}x$
16. $\displaystyle\int_0^{\infty} \frac{1}{(x^4+1)^2}\,\mathrm{d}x$
17. $\displaystyle\int_1^{\infty} \frac{1}{x+nx^n}\,\mathrm{d}x$, where $n>1$.
18. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{x}{\tan(x)}\,\mathrm{d}x$

19. $\displaystyle\int_0^{2025} \left(\frac{x}{\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}} - \sqrt{x-\sqrt{x-\sqrt{x-\cdots}}} \right) \mathrm{d}x$

20. $\displaystyle\int_0^{\frac{\pi}{2}} \int_0^1 \exp\left(t + t^{\tan(\theta)}\right) \mathrm{d}t \, \mathrm{d}\theta$

21. $\displaystyle\int_0^1 \cdots \int_0^1 \max(x_1, \cdots, x_{2025}) \, \mathrm{d}x_1 \cdots \mathrm{d}x_{2025}$

22. $\displaystyle\int_0^\infty \left(\frac{x}{(2^x-1)^2} - \frac{4x}{(4^x-1)^2} \right) \mathrm{d}x$

23. $\displaystyle\int_1^e \left(\sqrt{\ln(x)} + \frac{1}{e-1} \exp\left(\left(\frac{x-1}{e-1} \right)^2 \right) \right) \mathrm{d}x$

24. $\displaystyle\int_0^\infty \frac{x^2 + \frac{2^{-\frac{x}{6}}}{7} - \frac{2^{-\frac{x}{7}}}{6}}{(2^{-\frac{x}{7}} + 36x^2)(2^{-\frac{x}{6}} + 49x^2)} \mathrm{d}x$

25. $\displaystyle\int_0^{25} \sqrt{1 + x\sqrt{1 + (x+1)\sqrt{1 + (x+2)\sqrt{\cdots}}}} \mathrm{d}x$

26. $\displaystyle\int_0^1 \frac{x(x-1)}{(x+1)\ln(x)} \mathrm{d}x$

27. $\displaystyle\int \frac{2^x}{(1+\sqrt{3})^x + (2+\sqrt{3})^x} \mathrm{d}x$

28. $\displaystyle\int_{-90}^{90} \mathrm{mrt}(x^3 - (3+a)x^2 + (4+2a)x - (2+a)) \, \mathrm{d}a$, where $\mathrm{mrt}(f(x))$ is 0 if $f(x)$ has less than two distinct real roots, and otherwise is the smallest distance between distinct real roots of $f(x)$.

29. $\displaystyle\int_0^{124} (f(x))^2 \, \mathrm{d}x$, where $f : [0, \infty) \to \mathbb{R}$ satisfies $(f(x))^3 + xf(x) = 125$ for all $x \in [0, \infty)$.

30. $\displaystyle\int_1^2 \frac{\tan^{-1}(1+x)}{x} \mathrm{d}x$